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Devices, systems, and methods for an instrument accessory

Abstract

The present disclosure, in its various aspects, is directed to instrument accessory devices, implementation methods, and related delivery systems. Embodiments according to the present disclosure, including as described herein, may increase the effectiveness and efficiency of endoscopy procedures, such as ERCP. In one example, an embodiment includes an instrument accessory device with an expandable member, the device configured to receive an instrument through an instrument lumen, wherein the expandable member comprises an aperture on the inner surface and outer surface, and the outer surface is in fluid communication with the inner surface.

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Background/Summary

PRIORITY (1) The present application is a non-provisional of and claims the benefit of priority under 35 USC § 119 to, U.S. Provisional Application Ser. No. 62/987,482, filed Mar. 10, 2020, the disclosure of which is herein incorporated by reference in its entirety.

FIELD

(1) The present disclosure pertains to medical devices. More particularly, the present disclosure pertains to instrument accessory devices and related systems and methods, particularly as may increase the effectiveness and efficiency of endoscopy procedures.

BACKGROUND

(2) Endoscopes, for example, are used in medical procedures to examine and treat conditions within the digestive tract. Endoscopic Retrograde Cholangiopancreatography (ERCP) is used to examine and treat issues in the common bile duct and pancreatic ducts. In some procedures, cannulation of the bile duct can be difficult, with movement in the duodenum creating positioning and stabilization of the endoscope hard to achieve. Peristaltic movements in the duodenum can require medicating the patient to suppress such movements. Further, the tightly contracted musculature of the duodenal papilla requires high levels of precision to maneuver endoscope through the papilla opening. Consequently, the effectiveness and efficiency of the procedure may become compromised, and the inability to cannulate the common bile duct may result in a failed ERCP.

(3) It is with the above considerations in mind that the improvements of the present disclosure may be useful.

SUMMARY

(4) The present disclosure, in its various aspects, is directed generally to instrument accessory devices, implementation methods, and related systems. Embodiments according to the present disclosure, including those described herein, may increase particularly the effectiveness and efficiency of procedures used for the examination and treatment of conditions within the body, e.g., cannulation of the bile duct during ERCP.

(5) In an aspect, an instrument accessory device may include an expandable member having a proximal end and a distal end extending along a longitudinal axis, an inner layer having an inner surface defining an inner diameter of the expandable member, and an outer layer disposed about the inner layer and having an outer surface defining an outer diameter of the expandable member. The inner surface may define an instrument lumen extending from the proximal end toward the distal end of the expandable member. The instrument lumen may be configured to receive an instrument therethrough. The expandable member may comprise an aperture that extends radially outward from the instrument lumen to an exterior of the expandable member beyond the outer diameter. The instrument lumen may be in fluid communication with the exterior of the expandable member via the aperture. Moreover, the aperture may provide an unobstructed view between the instrument lumen and the exterior of the expandable member.

(6) In various embodiments described here or otherwise, the instrument lumen may be configured to receive an instrument. The instrument may be an endoscope or a duodenoscope. The instrument

lumen may be configured to maintain frictional contact with the instrument when the expandable member is in a collapsed condition. The inner layer and the outer layer may define inner and outer walls of a chamber that may be pressurized. The instrument accessory device may include an inflation lumen that extends through the outer layer of the expandable member and into the chamber. The inner layer may comprise a material configured to maintain frictional contact with an instrument. The outer layer may comprise a compliant elastic material.

(7) In an aspect, an endoscopic system may comprise a duodenoscope. The system may include an endoscope accessory device disposable about the instrument. The endoscope accessory may comprise an expandable member having a proximal end, a distal end, a longitudinal axis, an inner layer having an inner surface defining an inner diameter, and an outer layer disposed about the inner layer and having an outer surface defining an outer diameter. The instrument lumen may extend from the proximal end toward the distal end of the expandable member. The instrument lumen may be configured to receive an instrument therein. The expandable member may comprise an aperture that extends radially outward from the instrument lumen to an exterior of the expandable member such that the instrument lumen is in fluid communication with the exterior of the expandable member the via the aperture. The instrument lumen of the expandable member may be configured to slidably receive at least a distal portion of the duodenoscope.

(8) In various embodiments described here or otherwise, the expandable member may be configured to maintain frictional contact with the endoscope. The inner surface may comprise a body (e.g., a defined structure). The system may comprise a vacuum source. The expandable member may be placed in an expanded condition, stabilizing an instrument within a patient. The expandable member may be pneumatically, electrically, or mechanically expandable. The system may further comprise a fluid which is injected into the expandable member.

(9) In an aspect, a method of performing an endoscopy may comprise connecting an instrument accessory device to an instrument, the instrument accessory device comprising an expandable member and an instrument lumen. The expandable member may comprise an aperture that extends radially outward from the instrument lumen to an exterior of the expandable member such that the instrument lumen is in fluid communication with the exterior of the expandable member the via the aperture. The instrument and instrument accessory device may be inserted into a patient. The expandable member may be expanded within the patient to an expanded state.

(10) In various embodiments described here or otherwise, the method may also comprise applying a vacuum with the instrument. The expandable member may be expanded within the patient expanding an outer diameter such that the expandable member engages a patient lumen and stabilizes the instrument within the patient. Expanding the expandable member may comprise injecting a gas or liquid into the expandable member. The outer surface may be expandable to fit a body cavity. The instrument may comprise a duodenoscope. The instrument accessory device may comprise an inflation lumen extending from the proximal end of the instrument accessory device to an interior chamber of the expandable member. The aperture may be positioned such that the instrument interacts with a body cavity of the patient. The expandable member may be collapsed within the patient

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting examples of the present disclosure are described by way of example with reference to the accompanying figures, which are schematic and not intended to be drawn to scale. In the figures, each identical or nearly identical component illustrated is typically represented by a single numeral. For purposes of clarity, not every component is labeled in every figure, nor is every component of each embodiment of the disclosure shown where illustration is not necessary to

allow those of skill in the art to understand the disclosure. In the figures:

(2) FIG. 1 illustrates an exemplary endoscope of a type described in an embodiment of the present disclosure.

(3) FIG. 2A illustrates a partial cross-sectional view of an endoscope system with an instrument accessory device in an expanded configuration within a duodenum and an exemplary endoscope, such as the endoscope of FIG. 1, extended therethrough, in accordance with an embodiment of the present disclosure.

(4) FIG. 2B illustrates a partial cross-sectional view of an endoscope system with an instrument accessory device, such as the device of FIG. 2A, in a collapsed configuration within a body and an exemplary endoscope, such as the endoscope of FIG. 1, extended therethrough, in accordance with an embodiment of the present disclosure.

(5) FIG. 3 illustrates a side view of an instrument accessory device in an expanded configuration, in accordance with an embodiment of the present disclosure.

(6) FIG. 4A illustrates a longitudinal partial cross-sectional view of the instrument accessory device of FIG. 3, in a collapsed configuration, with an inserted endoscope.

(7) FIG. 4B illustrates a longitudinal partial cross-sectional view of the instrument accessory device of FIG. 3, in an expanded configuration, with an inserted endoscope.

(8) FIG. 5A illustrates a longitudinal partial cross-sectional view of the instrument accessory device of FIG. 4A, in a collapsed configuration.

(9) FIG. 5B illustrates a longitudinal partial cross-sectional view of the instrument accessory device of FIG. 4B, in an expanded configuration.

(10) FIG. 6A illustrates a radial cross-sectional view of an instrument accessory device in a collapsed configuration.

(11) FIG. 6B illustrates a radial cross-sectional view of an instrument accessory device in an expanded configuration.

(12) It is noted that the drawings are intended to depict only typical or exemplary embodiments of the disclosure. Accordingly, the drawings should not be considered as limiting the scope of the disclosure. The disclosure will now be described in greater detail with reference to the accompanying drawings.

DETAILED DESCRIPTION

(13) Various embodiments according to the present disclosure are described below. As used herein, “proximal end” refers to the end of a device that lies closest to the medical professional along the device when introducing the device into a patient, and “distal end” refers to the end of a device or object that lies furthest from the medical professional along the device during implantation, positioning, or delivery.

(14) As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

(15) It is noted that references in the specification to “an embodiment”, “some embodiments”, “other embodiments”, etc., indicate that the embodiment described may include one or more particular features, structures, and/or characteristics. However, such recitations do not necessarily mean that all embodiments include the particular features, structures, and/or characteristics. Additionally, when particular features, structures, and/or characteristics are described in connection with one embodiment, it should be understood that such features, structures, and/or characteristics may also be used in connection with other embodiments whether or not explicitly described unless clearly stated to the contrary.

(16) The detailed description should be read with reference to the drawings, which are not necessarily to scale, depict illustrative embodiments and are not intended to limit the scope of the invention.

(17) Embodiments of the present disclosure may include an instrument accessory as a device for use with an instrument. An instrument may be an endoscope, duodenoscope, colonoscope, bronchoscope, gastroscope, ureterscope, catheter, or the like. The device may include various components and configurations. Embodiments of this disclosure may comprise a medical device system. A medical device system may include an instrument, a handle, an instrument accessory, and/or a source of inflation fluid, or the like. Embodiments of the devices and systems may be used to fill a body lumen around the instrument, during an endoscopic procedure, to enable the operator to stabilize the instrument within the body. Various embodiments described herein comprise an instrument accessory device having an expandable member about an instrument lumen that can slidably receive an instrument. The expandable member once expanded within the body lumen may inhibit movement of the endoscope within the body lumen. In some embodiments, the inflation fluid may be a gas or a liquid. In some embodiments, the body lumen may include a lumen, organ, vessel, passage, or the like, within, e.g., the digestive system, or the like.

(18) Endoscopic Retrograde Cholangiopancreatography (ERCP) is used to examine and treat issues in the common bile duct and pancreatic ducts. An endoscope is introduced into the patient via the mouth, through the stomach and advanced through the small intestine. The endoscope may be used to access the ampulla of Vater in order to reach the pancreatic and bile ducts. The positioning of the endoscope relative to the duodenal papilla is key to the success of the procedure, and can be made difficult by movement of the duodenum.

(19) For various uses of endoscopes or other instruments, within various body lumens and for various purposes, such as described above, embodiments of the instrument accessory devices, systems and methods of the present disclosure may be utilized to enable the physician to perform procedures with greater precision, accuracy, and ease than without the accessory devices.

(20) An instrument accessory device may include an expandable member. An expandable member may include an inner layer having an inner surface which may be a non-elastic material such as Nylon, PET, PTFE, Pebax®. An expandable member may include an outer layer having outer surface which may be an elastic material such as a compliant balloon, a mesh, scaffold, a braid, or the like. An expandable member may be mechanically actuated, electrically actuated, pneumatically actuated, inflated, or the like. An expandable member may transition from a collapsed configuration to an expanded configuration to occlude, stretch, establish patency, or maintain patency of a body lumen.

(21) An instrument accessory device may have a fluid inlet that may extend through the outer surface of an expandable member. The fluid inlet may supply and remove an inflation fluid for inflating, or expanding and un-expanding or collapsing the expandable member.

(22) In one method of performing an endoscopy, a distal end of an endoscope may be advanced through the small intestine via the mouth of a patient. Once inside the patient, the digestive system can be visually examined and the pancreatic and bile ducts may be accessed using an instrument passed through a working channel from a handle at the proximal end of the endoscope that remains outside of the body.

(23) Referring to FIG. 1, an embodiment of a distal end of an endoscope of the type described in the present disclosure is illustrated. The endoscope **200** is one example of an instrument that can be used with an instrument accessory device described further herein. The endoscope **200** comprises a distal end **103** and a proximal end **104** with a lumen or working channel extending therethrough. A handle **105** at the proximal end **104** may be operated by a medical professional to manipulate the endoscope **200**. The endoscope **200** may include cuts or channels **106** along a wall **107** of the insertion portion of the endoscope **200** in order to facilitate movement and flexibility within a patient, e.g., by operation of steering knobs at the handle **105**.

(24) Referring to FIGS. 2A and 2B, a view of an endoscopic system within the body is illustrated according to an embodiment of the present disclosure, which includes an instrument accessory device **102** slidably disposed around an endoscope **200** (e.g., the endoscope **200** of FIG. 1). The

distal end **103** of the endoscope **200** and instrument accessory device **102** may be advanced through the small intestine **100**, with the proximal end of the endoscope **200** remaining outside the body. When the endoscope **200** is being advanced through the body, the instrument accessory device remains disposed about the endoscope **200** by maintaining frictional contact with the endoscope **200**. Once the endoscope **200** is in position, the expandable member **202** of the instrument accessory device **102** may be expanded, and a catheter **108** extended through the aperture **210**. The expandable member **202** of the device may substantially fill a cross-section of the body lumen **109** while in the expanded configuration. The expandable member **202** may be transitioned from a collapsed configuration to an expanded configuration by supply of an inflation fluid through a fluid inlet extending through the outer layer of the expandable member **202**. The body lumen **109**, e.g., small intestine, duodenum, etc., may be substantially occluded by the expandable member **202** when the expandable member **202** is in an expanded state. The endoscope **200** may be immovable within the instrument lumen **207**.

(25) Referring to FIG. 3, an instrument accessory device **102** is illustrated according to embodiments of the present disclosure. The instrument accessory device **102** includes an expandable member **202** having a proximal end **201**, a distal end **203**, and a longitudinal axis **205**. An inner layer **206** having inner surface defines an inner diameter of the expandable member **202**, extending from the proximal end **201** towards the distal end **203**. The inner layer **206** defines an instrument lumen **207** having an inlet **221** at the proximal end **201** and the instrument lumen **207** extending towards the distal end **203**. The instrument lumen **207** is configured to receive an instrument, e.g., the endoscope of FIG. 1, extendable through the instrument lumen **207**. A fluid inlet **214** extends from the proximal end **201** of the expandable member **202**.

(26) An outer layer **204** having outer surface defining an outer diameter of the expandable member **202** extends between the proximal **201** and distal **203** ends along a longitudinal axis **205**. The longitudinal axis **205** of the instrument lumen **207** and the longitudinal axis **205** of the expandable member **202** may be coextensive with each other, as shown, or may be radially offset from each other (not shown). The instrument lumen **207** and the expandable member **202** may each have a center axis as the longitudinal axis **205** that may be coextensive or radially offset from each other. The accessory devices shown in FIGS. 3, 4A, and 4B include fluid inlet **214** having a proximal end (not shown), a distal end, and an inflation lumen therethrough. The distal end of the fluid inlet **214** extends to the outer layer **204** of the expandable member **202**. An inner lumen of the fluid inlet **214** is in fluid communication with an interior chamber of the expandable member **202**. An aperture **210** is positioned between the proximal end **201** and the distal end **203** of the expandable member **202** and extends radially outward from the instrument lumen **207** to an exterior of the expandable member. The inner surface of the inner layer **206** of the expandable member **202** maintains frictional contact with an instrument such as an endoscope **200**. The inner layer **206** may be formed from a tube of a material, e.g., silicone, latex, or the like, which does not substantially change in dimension. The inner layer **206** is adhered to the outer layer **204** at the proximal **201** end of the expandable member **202**. The inner layer **206** is also adhered to the outer layer **204** around the aperture **210**. In some embodiments, the inner layer **206** is in the form of a tube having a lateral opening, with the material of the outer layer **204** extending into the aperture **210** and forming a sidewall of the aperture **210**. The adhered surfaces may be achieved by an adhesive, a tape, flowed material, welding (e.g., laser, ultrasonic, hot jaw thermal, etc.), melt/re-melt flow processes, or the like. The adhered layers create a substantially tight fluid seal between the inner layer **206** and the outer layer **204** of the expandable member **202** such that an inflation fluid may be supplied through the inflation lumen of the fluid inlet **214** to transition the expandable member between a collapsed configuration like that of FIGS. 4A, 5A, and 6A and an expanded configuration like that of FIGS. 4B, 5B, and 6B. The inner layer **206** and outer layer **204** define the instrument lumen which can be pressurized by supplying fluid through the fluid inlet **214** to achieve the expanded configuration of FIGS. 4B, 5B, and 6B. The expandable member **202** in the expanded configuration is illustrated in

an overall substantially cylindrical shape, but the expandable member **202** may be other shapes such as, e.g., ellipse, a sphere, a combination thereof, or the like. An endoscope **200** extending through the instrument lumen **207** of the expandable member **202** such that the distal end **212** of the endoscope **200** is in contact with the distal end **203** of the expandable member **202** may be used to examine a body lumen, as the inner layer **206** remains in frictional contact with the instrument. The working channel **208** of the endoscope **200** aligns with the aperture **210** of the expandable member **202**. By transforming the expandable member **202** into the expanded configuration in a body lumen with the aperture **210** positioned toward the papilla, the expandable member **202** prevents movement of the endoscope **200** while access is made to the papilla.

(27) The instrument accessory device **202** is illustrated in FIG. 3 such that it is configured to receive an endoscope **202** through the proximal end **201** toward the distal end **203**. The expandable member **202** of the instrument accessory device **102** is shown expanded, which is the configuration where the instrument lumen **207** may slidably receive the endoscope **200**.

(28) In various embodiments, an expandable member may have an expanded configuration and a collapsed configuration. An expandable member may comprise an outer layer forming an outer surface and an inner layer forming an inner surface. An outer layer may comprise a variety of compliant, semi-compliant, or non-compliant materials. These materials may comprise silicone, latex, polyurethane, rubber, isobutylene or the like. The thickness of a wall of the outer layer may vary with the material and may relate to the outer diameter of the outer layer in the collapsed and the expanded configuration. An inner layer may comprise a variety of semi-compliant, or non-compliant materials. These materials may comprise silicone, latex or the like. The thickness of a wall of the inner layer may vary with the material. The inner surface and outer surface may not expand to the same degree. An expandable member may be expanded and collapsed or otherwise stretched once or a plurality of times to increase its elasticity prior to use within a patient, which may improve a symmetrical inflation of the expandable member and may improve the centering mechanics of the expandable member. An expandable member may be expanded via a supply of an inflation fluid through one or more fluid inlets. The same fluid inlet may be used to expand and deflate the expandable member. Alternatively, a supply fluid inlet and a return fluid inlet may be intermittently or continuously used to circulate inflation fluid through an expandable member. A continuous flow of inflation fluid through a supply fluid inlet and a return fluid inlet may substantially maintain a desired pressure of the inflation fluid within the expandable member, or the inflation fluid may be heated and circulated to maintain a desired temperature with the expandable member.

(29) In various embodiments, a vacuum may be applied by the instrument. The vacuum may pull the wall of the body lumen into the aperture. Where the aperture surrounds a duodenal papilla, the vacuum may draw open the papilla.

(30) In various embodiments, the distal end of the instrument may or may not extend past the distal end of the instrument accessory device. If the distal end of the instrument does not extend past the distal end of the instrument accessory device, the expandable member may surround the distal end of the instrument.

(31) In various embodiments, the inner surface may expand to a lesser degree than the outer surface. The inner and outer surfaces may comprise a polymeric material.

(32) In various embodiments, the instrument accessory device may be disposable. In alternate embodiments, the instrument accessory device may be reusable.

(33) In various embodiments, an inflation fluid supplied through a fluid inlet may include saline, water, CO₂, dilute contrast media, or the like.

(34) In various embodiments, a method of performing an endoscopy may include placing an expandable instrument accessory device on an instrument. The instrument and instrument accessory device are together inserted into a patient. The expandable member can be expanded within the patient, dilating the patient's body lumen, e.g., the duodenum, the intestines, or the like.

The instrument accessory device increases in diameter when the expandable member is expanded, inhibiting movement of both the expandable instrument accessory device and the instrument within the body lumen. Expanding the expandable member allows the instrument to remain still while using the instrument to examine other areas of the body. This can occur due to the outer diameter dilating the body lumen, preventing movement of the device. The instrument may be any device used to perform an endoscopy, e.g., ERCP or the like.

(35) Variations, modifications, and other implementations of the present disclosure in addition to the various embodiments described herein will occur to those of ordinary skill in the art.

Accordingly, the present disclosure is to be defined not by the preceding illustrative description but instead by the following claims:

Claims

1. An endoscopic system, comprising: a duodenoscope in fluid communication with a vacuum source; and an endoscope accessory device disposable about the duodenoscope, comprising: an expandable member having a proximal end, a distal end, a longitudinal axis, an inner layer having an inner surface defining an inner diameter of the expandable member, and an outer layer having an outer surface defining an outer diameter of the expandable member disposed about the inner layer, the inner surface defining an instrument lumen extending from the proximal end toward the distal end of the expandable member, the instrument lumen configured to receive an instrument therethrough; wherein the inner layer and the outer layer define inner and outer walls of a chamber that can be pressurized; wherein the expandable member comprises an aperture that extends radially outward from the instrument lumen to an exterior of the expandable member such that the instrument lumen is in fluid communication with the exterior of the expandable member via the aperture; wherein the expandable member is configured such that, when the device is disposed within a body lumen, a wall of the body lumen can be pulled into the aperture by a vacuum applied from the vacuum source by the duodenoscope; and wherein the instrument lumen of the expandable member is configured to slidingly receive at least a distal portion of the duodenoscope.
 2. The system of claim 1, wherein the expandable member is configured to maintain frictional contact with the duodenoscope.
 3. The system of claim 1, wherein the inner surface comprises a body.
 4. The system of claim 1, wherein the system further comprises a vacuum source.
 5. The system of claim 1, wherein when the expandable member is in an expanded condition, the endoscope is stabilized within a patient.
 6. The system of claim 1, wherein the expandable member is pneumatically, electrically, or mechanically expandable.
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